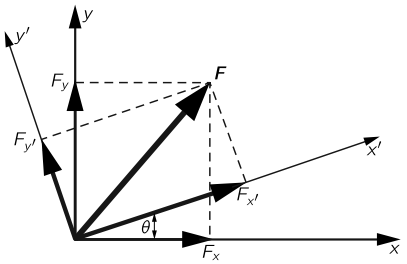

Figure 1-3 Components of a force in the two systems.

It is interesting to note an accident of sorts, which is of extreme importance: the formulas (1.5) and (1.6), for coordinates of P and components of F, respectively, *are of identical form*.

As before, Newton's laws are assumed to be true in Joe's system, and are expressed by equations (1.1). The question, again, is whether Moe can apply Newton's laws—will the results be correct for his system of rotated axes? In other words, if we assume that Eqs. (1.5) and (1.6) give the relationship of the measurements, is it true or not true that

$$\begin{aligned} m(d^2x'/dt^2) &= F_{x'}, \\ m(d^2y'/dt^2) &= F_{y'}, \\ m(d^2z'/dt^2) &= F_{z'}? \end{aligned} \tag{1.7}$$

To test these equations, we calculate the left and right sides independently, and compare the results. To calculate the left sides, we multiply equations (1.5) by m, and differentiate twice with respect to time, assuming the angle θ to be constant. This gives

$$\begin{aligned} m(d^2x'/dt^2) &= m(d^2x/dt^2)\cos\theta + m(d^2y/dt^2)\sin\theta, \\ m(d^2y'/dt^2) &= m(d^2y/dt^2)\cos\theta - m(d^2x/dt^2)\sin\theta, \\ m(d^2z'/dt^2) &= m(d^2z/dt^2). \end{aligned} \tag{1.8}$$

We calculate the right sides of equations (1.7) by substituting equations (1.1) into equations (1.6). This gives